

UChicago REU Notes Week 3

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Contents

1 Operad: Alternative Definition and Relation to Monad (ND, a bunch of definition problems)

For the last section, we've seen an associative non- Σ operad, to produce the version in operad (equipped with symmetric group actions), we need to consider another version. But for this, I think there are some technicalities in the category theory, so we'll assume $\mathcal{V}$ is monoidally cocomplete.

An alternative assumption can be made that internal hom functor exists. Out of curiosity, are the two equivalent definition? Also, are there any symmetric monoidal category where it doesn't commute with coproduct?

Example 1.1. Let $\mathcal{M}(j) := \kappa[\Sigma_j] := \coprod_{\psi \in \Sigma_j} \kappa_{\psi}$, coproduct of $\#\Sigma_j$ copies of κ (each κ_{ψ} is the unit κ , subscript is for labeling convenience). It has a natural right Σ_j -action, where for each $\sigma \in \Sigma_j$, the identity $\kappa_{\psi\sigma^{-1}} \xrightarrow{\sim} \kappa_{\psi}$ induces an isomorphism on coproduct $\tilde{\sigma} \in \text{Aut}(\mathcal{M}(j))$:

$$\begin{array}{ccc} \kappa_{\psi} & \xrightarrow{\sim} & \kappa_{\psi\sigma} \\ \downarrow \iota_{\psi} & & \downarrow \iota_{\psi\sigma} \\ \coprod_{\psi \in \Sigma_j} \kappa_{\psi} & \xrightarrow[\exists! \tilde{\sigma}]{} & \coprod_{\psi \in \Sigma_j} \kappa_{\psi\sigma} \end{array}$$

This construction enforces $(\sigma\tilde{\sigma}') = \tilde{\sigma}' \circ \tilde{\sigma}$, which is a group homomorphism $\Sigma_j^{\text{op}} \rightarrow \text{Aut}_{\mathcal{V}}(\mathcal{M}(j))$.

There is a natural unit morphism $\eta : \kappa \rightarrow \kappa[\Sigma_1] = \kappa$ via identity.

Also, each identity morphism $\kappa_{\psi} \rightarrow \kappa$ induces a trivialization $t : \kappa[\Sigma_j] \rightarrow \kappa$. Combining the action $\psi \in \Sigma_k$ (by identifying concatenation of blocks $(j_{\psi^{-1}(1)}, \dots, j_{\psi^{-1}(k)}) = \{1, \dots, j\}$ in order, also inducing inclusion $\Sigma_{j_s} \hookrightarrow \Sigma_j$), we get the following morphism:

$$\kappa_{\psi} \otimes \left(\bigotimes_{s=1}^k \kappa[\Sigma_{j_s}] \right) \xrightarrow{\sim} \bigotimes_{s=1}^k \kappa[\Sigma_{j_{\psi^{-1}(s)}}] \xrightarrow{t^{k-1}} \kappa[\Sigma_{j_{\psi^{-1}(s)}}] \rightarrow \kappa[\Sigma_j]$$

Running through all $\psi \in \Sigma_k$, this induces a morphism $\gamma : \coprod_{\psi \in \Sigma_k} \left[\kappa_{\psi} \otimes \left(\bigotimes_{s=1}^k \kappa[\Sigma_{j_s}] \right) \right] \rightarrow \kappa[\Sigma_j]$, and use the fact $\otimes$ commutes with colimits, this is equivalent to a morphism $\gamma : \kappa[\Sigma_k] \otimes \left(\bigotimes_{s=1}^k \kappa[\Sigma_{j_s}] \right) \rightarrow \kappa[\Sigma_j]$.

This is helpful when defining an alternative definition.

Use Σ to denotes the category of finite sets, each $n \in \mathbb{N}$ has $\mathbf{n} := \{1, \dots, n\}$ (and $\mathbf{0}$ denotes the emptyset), with homsets being all possible isomorphisms (so $n \neq m$ there's no morphism between them, while the

endomorphism is consists of permutations). Each object $\mathbf{n}$ has a canonical Σ_n -action. This is called a *permutation category*.

Definition 1.2. We define a Σ -object in the monoidal category $\mathcal{V}$ to be a functor $F : \Sigma^{\text{op}} \rightarrow \mathcal{V}$, which each $F(n)$ is equipped with a natural right Σ_n -action.

A morphism between Σ -objects will be natural transformations, which by definition each morphism at $\mathbf{n}$ is a Σ_n -equivariant morphism, as all $\sigma \in \Sigma_n$ (as a right action, equivalently an isomorphism) satisfies the following condition:

$$\begin{array}{ccc} F(\mathbf{n}) & \xrightarrow{\sigma} & F(\mathbf{n}) \\ \mu_n \downarrow & & \downarrow \mu_n \\ G(\mathbf{n}) & \xrightarrow{\sigma} & G(\mathbf{n}) \end{array}$$

this forms a *category of Σ -objects* for $\mathcal{V}$.

Now, consider the following operation:

Definition 1.3. Given F, G two Σ -objects of $\mathcal{V}$, define the *substitution product* $G \circ F : \Sigma^{\text{op}} \rightarrow \mathcal{V}$ as follow:

$$G \circ F(\mathbf{j}) := \coprod_{j_1 + \dots + j_k = j} G(\mathbf{k}) \otimes_{\kappa[\Sigma_k]} \left[\left(\bigotimes_{r=1}^k F(\mathbf{j}_r) \right) \otimes_{\kappa[\Sigma_{j_1 \times \dots \times j_k}]} \kappa[\Sigma_j] \right]$$

Let's assume we know these are associative due to the coequalizer property :)

But roughly speaking, an operad is a Σ -object, that is also a monoid under the above product. (There are more details to work out...)

Talking about monads relevant to an operad $\mathcal{C}$ over $\mathcal{V}$, we define it as follow:

Definition 1.4. Define a functor $C : \mathcal{V} \rightarrow \mathcal{V}$ as follow:

$$\coprod_{j \geq 0} \mathcal{C}(j) \otimes \kappa[\Sigma_j] \otimes X^j \rightrightarrows \coprod_{j \geq 0} \mathcal{C}(j) \otimes X^j \xrightarrow{\text{coeq}} C(X)$$

Where given any $\psi \in \Sigma_j$, define the top morphism using $\mathcal{C}(j) \otimes \kappa_\psi \otimes X^j \xrightarrow{\sigma \otimes \text{id}} \mathcal{C}(j) \otimes X^j$ (action on $\mathcal{C}(j)$), which induces left action morphism $\mathcal{C}(j) \otimes \kappa[\Sigma_j] \otimes X^j \xrightarrow{\ell} \mathcal{C}(j) \otimes X^j$, then using coproduct universality to induce the morphism.

The bottom morphism is defined using $\mathcal{C}(j) \otimes X^j \xrightarrow{\text{id} \otimes \sigma^{-1}} \mathcal{C}(j) \otimes X^j$ (by sending the i th index X to the $\sigma^{-1}(i)$ th index X), then induce the coproduct morphism using similar logic (and label it as $\mathcal{C}(j) \otimes \kappa[\Sigma_j] \otimes X^j \xrightarrow{r} \mathcal{C}(j) \otimes X^j$). This is formally saying: A right action on $\mathcal{C}(j)$ transfers to an inverse left action on X^j .

Given a morphism $\psi : X \rightarrow Y$, the morphism $\text{id} \otimes \text{id} \otimes \psi^j : \mathcal{C}(j) \otimes \kappa[\Sigma_j] \otimes X^j \rightarrow \mathcal{C}(j) \otimes \kappa[\Sigma_j] \otimes Y^j$, and $\text{id} \otimes \psi^j : \mathcal{C}(j) \otimes X^j \rightarrow \mathcal{C}(j) \otimes Y^j$ bot satisfies the following:

$$\begin{array}{ccc} \mathcal{C}(j) \otimes \kappa[\Sigma_j] \otimes X^j & \xrightarrow{\ell, r} & \mathcal{C}(j) \otimes X^j \\ \text{id} \otimes \text{id} \otimes \psi^j \downarrow & & \downarrow \text{id} \otimes \psi^j \\ \mathcal{C}(j) \otimes \kappa[\Sigma_j] \otimes Y^j & \xrightarrow{\ell, r} & \mathcal{C}(j) \otimes Y^j \end{array}$$

As including $\mathcal{C}(j) \otimes \kappa_\sigma \otimes X^j$, the top right route provides the morphism $\mathcal{C}(j) \otimes \kappa_\sigma \otimes X^j \xrightarrow{\sigma \otimes \text{id} \otimes \text{id}} \mathcal{C}(j) \otimes X^j \xrightarrow{\text{id} \otimes \psi^j} \mathcal{C}(j) \otimes Y^j$, equivalent to $\mathcal{C}(j) \otimes X^j \xrightarrow{\sigma \otimes \psi^j} \mathcal{C}(j) \otimes Y^j$, and by chasing through the coproduct property, we realize the following diagram commutes:

$$\begin{array}{ccc} \mathcal{C}(j) \otimes \kappa_\sigma \otimes X^j & \longrightarrow & \mathcal{C}(j) \otimes \kappa[\Sigma_j] \otimes X^j \\ \text{id} \otimes \text{id} \otimes \psi^j \downarrow & & \downarrow \text{id} \otimes \text{id} \otimes \psi^j \\ \mathcal{C}(j) \otimes \kappa_\sigma \otimes Y^j & \longrightarrow & \mathcal{C}(j) \otimes \kappa[\Sigma_j] \otimes Y^j \end{array}$$

Combining the two squares, do some diagram chasing, we realize the original square is commuting (with a pinch of coproduct property on $\kappa[\Sigma_j]$).

Finally, these induce the morphism between coproducts, and they of course commute with the group action by construction (as permuting ψ^j is equivalent to pre-permuting with ψ^j , or compose ψ^j with post-permuting). So, composing with the coequalizing morphism of Y , the desired diagram factors through coequalizing morphism of X , inducing a morphism $C(X) \rightarrow C(Y)$. Such induced morphism is functorial, hence C is indeed a functor.